RAJALAKSHMI ENGINEERING COLLEGE

Lathikka D 251001133-2025-IT

L2

Started on	Friday, 9 October 2026, 7:02 PM
State	Finished
Completed on	Friday, 9 October 2026, 7:03 PM
Time taken	58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
int x = *(const int *)a;
int y = *(const int *)b;
return (x > y) - (x < y);
}

int main() {
int n, arr[100000];
long long sum = 0;

scanf("%d", &n);

for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

qsort(arr, n, sizeof(int), compare);

for (int i = 0; i < n; i++) {
    sum += (long long)arr[i] * i;
}

printf("%lld\n", sum);

return 0;


}

```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔

	Input	Expected	Got	
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course